# Telamorph website case studies (anonymised)

Five project pages for telamorph.com, written from the two portfolio PDFs with client, brand and model names removed. Each one can go on its own page and then be linked from the founders' LinkedIn Featured sections. Go through the checks at the end before anything is published.

## Projects page intro

## 1. Complete composite body for a sports car

We engineered and industrialised a complete composite body for a sports car, built to the customer's exterior design. The 26 kg body has been fitted to about 15 cars and is ready for a third production series.

- 26 kg complete composite body
- Fitted to about 15 cars
- Fully documented and ready for a third series

### The brief

The customer supplied the design data for the exterior. The body had to meet defined requirements for dimensional accuracy, fit, durability, reliability and structural performance, and it had to be repeatable in low-volume production.

### What we did

- Engineering of all exterior panels and selected interior components
- Material selection, structural design and sizing
- Component interfaces, joining solutions and body-to-chassis mounting
- Design for manufacture
- Design and manufacture of all tooling
- The first complete prototype body, validated against the requirements
- Industrialisation and low-volume production

### How it is built

The body is a hybrid of carbon fibre and carbon-Kevlar, optimised for manufacturability and high structural stiffness. It is designed to carry the defined service loads without permanent deformation, at minimum structural weight.

### Result

A 26 kg complete composite body, fitted to approximately 15 vehicles. The project is fully documented and ready for a third manufacturing series.

| Material | Structure | Process | Scope |
|---|---|---|---|
| Hybrid carbon fibre and carbon-Kevlar | Selected multi-layer laminates | Vacuum infusion, CNC trimming, bonding, fitting, finishing | Complete body and selected interior components |

# 2. Composite monocoque and structure for a certified light aircraft

We developed and built the composite monocoque cockpit and structure for a certified light aircraft. After certification it went into series production, and it has been built for nearly 70 aircraft.

- Certified, then taken into series production
- Built for nearly 70 aircraft
- Several aircraft past 2,500 flight hours and more than five years in service

## The brief

The customer's engineering team needed a composite cockpit and structure that would pass certification testing and then go into series production. The structure had to carry the loads from the engine, the propeller and other highly loaded assemblies.

## What we did

- Contribution to the aerodynamic geometry, together with the customer's engineering team
- Structural architecture, including the load-bearing interfaces to the engine, propeller and other highly loaded assemblies
- Structural sizing and laminate definition
- Thermal design around high-temperature zones
- Assembly design and tolerance definition
- Design and manufacture of the tooling, and dimensional control
- The first complete prototype structure, built and assembled for certification testing
- Industrialisation and series production

## How it is built

Each component has a composite layup tailored to its own load case. Fibre orientation and laminate thickness vary locally within individual parts, and stiffening closing panels limit the deformation of lightly loaded aerodynamic surfaces. Honeycomb cores and reinforcement fabrics are chosen locally to balance stiffness, elasticity and weight.

## Result

A certified composite structure, taken into series production and built for nearly 70 aircraft. Several of these aircraft have passed 2,500 flight hours and more than five years in service.

| Material | Structure | Process | Scope |
|---|---|---|---|
| Carbon fibre, glass fibre, honeycomb cores | Mass- and load-optimised composite structure | Vacuum infusion, prepreg, RTM | Complete structure |

# 3. Radio-transparent radome insert for a UAV

We engineered and built a compact radome that sits in the composite outer skin of an unmanned aerial vehicle and gives its antenna a controlled non-conductive window. After first-article validation, the part went into low-volume production.

- Kevlar-Nomex sandwich window with controlled wall thickness
- Out-of-autoclave prepreg
- Taken into low-volume production

## The brief

The customer supplied the outer-mould-line geometry, the antenna envelope and the installation constraints. The radome had to keep the airframe's aerodynamic contour, give the antenna a non-conductive window and fit the surrounding structure.

## What we did

- Material selection, laminate and wall-thickness definition
- Peripheral flange and sealing design
- Structural checks for aerodynamic and vibration loads
- Manufacturing tolerances
- Joints, reinforcements and hardware placed outside the active RF window
- Preliminary electromagnetic assessment of the material system, laminate thickness and geometry
- Tooling and manufacturing route
- First article for dimensional inspection, airframe fit checks, sealing verification and customer-led RF testing

## How it is built

The active window is a Kevlar-Nomex sandwich with an epoxy laminate, made with controlled fibre orientation, resin content and wall thickness. Non-conductive reinforcement is limited to the attachment perimeter, and conductive fasteners and the surrounding carbon-fibre structure stay outside the RF aperture. A bonded peripheral flange takes the local installation loads, and the outer surface has a lightweight dielectric coating suited to the laminate and the operating

environment.

## Result

The first article kept the UAV's aerodynamic outer contour and gave a controlled non-conductive window for system-level antenna validation. The radome then went into low-volume production.

| Material | Structure | Process | Scope |
|---|---|---|---|
| Kevlar-Nomex sandwich, epoxy, dielectric coating | Controlled-thickness RF window, reinforced perimeter | Out-of-autoclave prepreg with Nomex core, CNC trimming, bonding, sealing, inspection | Complete radome insert |

# 4. Watertight composite enclosure for a compact underwater drone

We developed and built the pressure-resistant composite enclosure and internal support structure for a compact underwater vehicle, and defined how it is inspected and pressure-tested.

- Glass-fibre/epoxy shell with local carbon-fibre reinforcement
- Bonded ring frames and an equipment cradle that keep the propulsion and sensors aligned
- Staged leak and hydrostatic pressure testing

## The brief

The customer defined the external envelope and the internal equipment layout. The enclosure had to resist external pressure, stay sealed, take the propulsion unit and hold the batteries and control electronics in alignment, with access for maintenance.

## What we did

- Enclosure architecture and shell segmentation
- Laminate definition
- External-pressure and buckling assessment
- Sealing geometry and service access
- Reinforced interfaces for the propulsion unit
- Internal supports for the batteries and control electronics, designed for load transfer, alignment, assembly sequence and maintenance access
- Design and manufacture of the tooling
- The prototype, the dimensional inspection and a staged hydrostatic pressure-testing procedure

## How it is built

The enclosure is a moulded glass-fibre/epoxy shell with local carbon-fibre reinforcement in highly loaded regions. Laminate thickness and fibre orientation vary around openings, mounting points and concentrated load paths. Bonded ring frames and a composite equipment cradle stabilise the shell and keep the enclosure, propulsion system and sensor interfaces aligned. The watertight joints use CNC-trimmed flanges and replaceable elastomeric seals, and metallic inserts are electrically isolated from the carbon reinforcement.

## Result

A compact, structurally integrated enclosure, verified through dimensional inspection, assembly checks and staged leak and pressure testing against the project requirements.

| Material | Structure | Process | Scope |
|---|---|---|---|
| Glass fibre, carbon fibre, epoxy | Pressure-bearing shell, bonded frames, equipment cradle | Vacuum infusion, CNC trimming, bonding, sealing, testing | Complete watertight enclosure |

# 5. Composite battery enclosure for an electric bicycle

We developed and built a compact composite enclosure that integrates a removable battery module into an electric bicycle frame. The design and tooling were prepared for repeatable small-series production.

- Flame-retardant epoxy system with insulating inner layers
- Removable battery behind a replaceable seal
- Design and tooling ready for small-series production

## The brief

The customer provided the battery envelope, the bicycle-frame geometry and the electrical interfaces. The enclosure had to take road vibration, handling loads and local impacts without passing concentrated loads into the battery module, keep conductive parts electrically isolated, and let the battery be removed for service.

## What we did

- Enclosure architecture and mounting concept
- Load transfer into the frame and internal battery supports
- Removable service access, sealing flanges, and cable and charging interfaces
- Tolerance definition and design for manufacture
- Electrical isolation around conductive composite layers and metallic inserts

- Thermal clearances, pressure equalisation and a controlled vent path, coordinated with the battery-system requirements
- Tooling, and prototype units for dimensional, assembly, sealing and functional checks

## How it is built

The enclosure uses a glass-fibre-rich laminate with local carbon-fibre reinforcement, a flame-retardant epoxy system and electrically insulating inner layers around the battery module. Bonded inserts are isolated from the carbon laminate. An overlapping cover flange and a replaceable elastomeric seal protect the battery and keep it serviceable, while internal supports and elastomeric pads locate the module and spread the loads across the enclosure. A membrane vent equalises pressure without opening a path for moisture.

## Result

The enclosure was integrated with the bicycle frame and electrical system. After checks of frame fit, fastener retention, gasket compression, connector access and basic water-ingress protection, the design and tooling were prepared for repeatable small-series manufacture.

| Material | Structure | Process | Scope |
|---|---|---|---|
| Glass fibre, carbon fibre, epoxy, local inserts | Sealed enclosure, reinforced mounts, isolated supports | Prepreg, CNC trimming, bonding, sealing, assembly | Complete battery enclosure |